

Deep Learning for Structural Corporate Finance Models: Solution and Estimation

Xingguang (Ray) Jin*

GitHub repository: <https://github.com/RayxgJIN/2025-Question-4>

1 Summary

This report develops a deep-learning framework to solve and estimate dynamic structural corporate finance models inspired by [Strebulaev and Whited \(2012\)](#), using global residual-minimization methods in the spirit of [Maliar et al. \(2021\)](#).

Main deliverables.

- **Baseline model (risk-free):** solve the investment problem with persistent productivity shocks by minimizing Bellman residuals using neural approximators for value and policy functions.
- **Risky-debt extension:** jointly learn equity value, capital/debt policies, and a reduced-form risky interest-rate schedule via Bellman + pricing residuals.
- **Estimation:** compare GMM vs. SMM estimation using moment matching on synthetic or empirical targets.
- **Bonus extension: CEO–Board block:** add effort/manipulation/risk choice and contract design; outline solution and identification.

References. [Strebulaev and Whited \(2012\)](#); [Maliar et al. \(2021\)](#); [Cronqvist et al. \(2024\)](#).

2 Part 1. Deep-learning solution of a corporate investment model

2.1 Baseline model setup ([Strebulaev and Whited \(2012\)](#), Sec. 3.1)

State. $s = (k, z)$ where $k \geq 0$ is capital and z is productivity.

Shock process (log AR(1)).

$$\ln z' = \rho \ln z + \varepsilon', \quad \varepsilon' \sim (\text{truncated}) \text{ Normal}, \quad (1)$$

with transition kernel $g(z' | z)$.

*Email: rayjinmath@gmail.com

Control and capital accumulation. The control is next-period capital $k' \geq 0$. Investment is

$$I = k' - (1 - \delta)k. \quad (2)$$

Profits and adjustment costs.

$$\pi(k, z) = z k^\theta, \quad 0 < \theta < 1, \quad (3)$$

$$\psi(I) = \frac{\phi_I}{2} I^2, \quad \phi_I > 0. \quad (4)$$

Notation fix: ϕ_I is the adjustment-cost parameter (not a neural-net parameter).

Bellman equation. Let $\beta = \frac{1}{1+r}$. The problem is

$$V(k, z) = \max_{k' \geq 0} \left\{ \pi(k, z) - \psi(k' - (1 - \delta)k) - (k' - (1 - \delta)k) + \beta \int V(k', z') dg(z' | z) \right\}. \quad (5)$$

2.2 (a) Maliar-style deep residual formulation

Neural approximations. Approximate:

$$V_{\theta_V}(k, z), \quad k'_{\theta_P}(k, z) \geq 0.$$

Define implied investment $I_{\theta_P}(k, z) = k'_{\theta_P}(k, z) - (1 - \delta)k$.

Bellman residual.

$$\mathcal{R}(k, z; \theta_V, \theta_P) = V_{\theta_V}(k, z) - \left[\pi(k, z) - \frac{\phi_I}{2} I_{\theta_P}(k, z)^2 - I_{\theta_P}(k, z) + \beta \mathbb{E}[V_{\theta_V}(k'_{\theta_P}(k, z), z') | z] \right]. \quad (6)$$

Monte Carlo expectation.

$$\mathbb{E}[\cdot | z] \approx \frac{1}{M} \sum_{m=1}^M (\cdot) \Big|_{z' = \exp(\rho \log z + \varepsilon^{(m)})}.$$

Training objective.

$$\min_{\theta_V, \theta_P} \mathbb{E}_{(k, z) \sim \mu} [\mathcal{R}(k, z; \theta_V, \theta_P)^2] + \lambda (\|\theta_V\|_2^2 + \|\theta_P\|_2^2), \quad (7)$$

where μ is the training distribution over states (mixture of broad random + on-policy samples recommended).

Constraint handling. Enforce $k' \geq 0$ by construction:

$$k'_{\theta_P}(k, z) = \text{softplus}(\text{NN}_{\theta_P}(k, z)), \quad \text{softplus}(x) = \log(1 + e^x),$$

optionally clipped to $[0, k_{\max}]$ for numerical stability.

2.3 Numerical configuration (risk-free baseline)

Please refer to Figure 1 and Table 1.

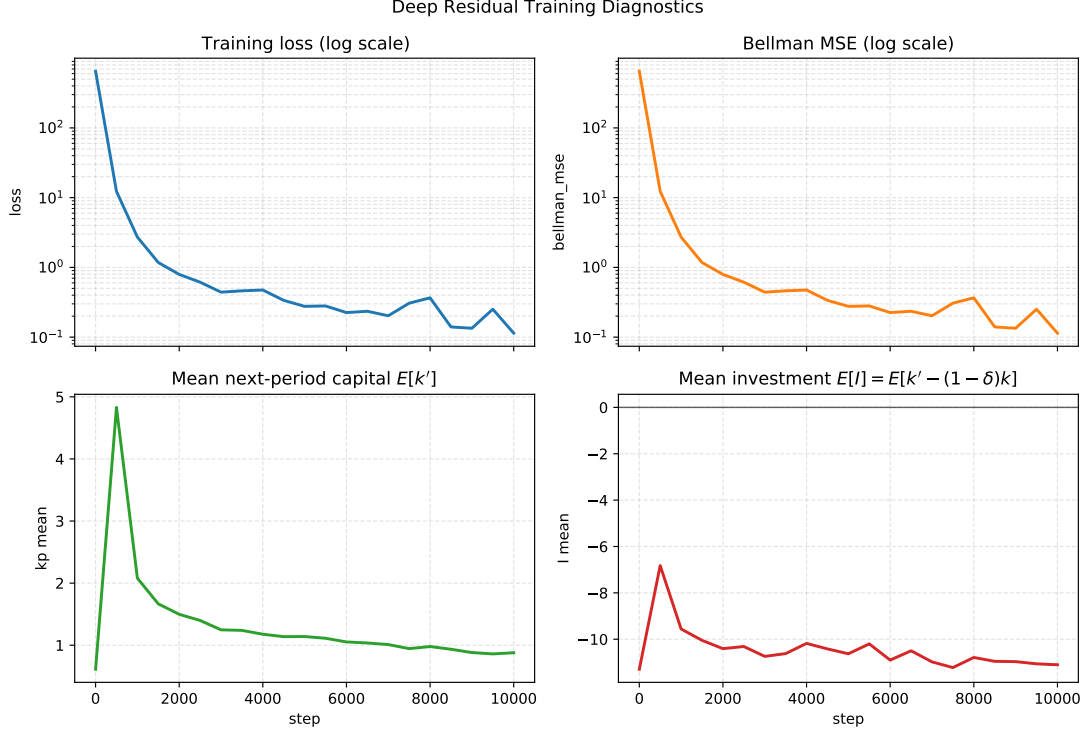


Figure 1: Training diagnostics. Top row: total loss and Bellman MSE (both on a log scale). Bottom row: means of next-period capital k' and investment $I = k' - (1 - \delta)k$.

Category	Setting / value
Risk-free rate	$r = 0.04$
Discount factor	$\beta = (1 + r)^{-1} \approx 0.961538$
Profit curvature	$\theta = 0.7$
Depreciation	$\delta = 0.15$
Adjustment cost	$\phi_I = 0.01$
Shock persistence	$\rho = 0.7$
Shock innovation s.d.	$\sigma_\varepsilon = 0.15$
State sampling (training)	log-uniform in bounded box
Value net V_{θ_V}	MLP (256, 256), tanh, linear output
Policy net k'_{θ_P}	MLP (256, 256), tanh, softplus output
Optimizer	Adam; lr 2×10^{-4} ; batch 4096
Training steps	10,000
MC draws	train 16; eval recommended 64
Stabilizers	grad clip 1.0; weight decay $\lambda = 10^{-8}$

Table 1: Baseline (risk-free) DRM settings.

2.4 Interpreting typical training patterns

- The loss of Bellman residual decreases sharply from 6.48×10^2 at step 1 to approximately 1.14×10^{-1} by step 10,000, indicating substantial improvement in satisfying the Bellman fixed point over the training distribution.
- The mean of k' initially jumps (reaching about 4.83 around step 500) and then declines, stabilizing near 0.9–1.1 by the end of training. In contrast, the mean investment I remains

strongly negative throughout training (approximately -7 to -11), suggesting persistent disinvestment on average in the sampled states.

- A persistently negative $E[I]$ can arise either (i) from an economically meaningful policy in which firms optimally run down capital under high leverage/low productivity, or (ii) from numerical artifacts such as policy clipping at bounds, non-stationary state sampling that places heavy mass on high- k states, or scaling issues that distort the optimizer’s incentives.

2.5 (b) Synthetic data and effectiveness metrics

Synthetic panel simulation. Given learned $(\hat{V}, \hat{k}')$, simulate for $t = 0, \dots, T - 1$:

$$\log z_{t+1} = \rho \log z_t + \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, \sigma_\varepsilon^2), \quad (8)$$

$$k_{t+1} = \hat{k}'(k_t, z_t), \quad (9)$$

$$I_t = k_{t+1} - (1 - \delta)k_t, \quad \pi_t = \pi(k_t, z_t). \quad (10)$$

Effectiveness definition. A solution is effective if it (i) yields small *out-of-sample* Bellman residuals, (ii) is close to one-step greedy improvement, and (iii) produces feasible and stable simulated paths.

Reportable metrics.

- Bellman residual RMSE: $\text{BRMSE} = \sqrt{\mathbb{E}[\mathcal{R}^2]}$ on a test set.
- Normalized BRMSE: $\sqrt{\mathbb{E}\left[\left(\mathcal{R}/(1 + |\hat{V}|)\right)^2\right]}$.
- Greedy gap: $\Delta(k, z) = \max_{k' \in \mathcal{K}} T(k, z; k') - T(k, z; \hat{k}'(k, z))$ (grid/line search).
- Simulated-path checks: moments of I_t/k_t , corner frequency $k_{t+1} \approx 0$, divergence/NaN checks.

Metric	Value
Bellman residual RMSE	7.8144×10^{-1}
Normalized BRMSE	7.9420×10^{-2}

Table 2: Baseline solution: effectiveness metrics (held-out evaluation).

2.6 (c) Key NN/training design issues

1. **Constraints:** enforce $k' \geq 0$ via parameterization; avoid dead gradients.
2. **Scaling:** use $(\log k, \log z)$ or standardized inputs.
3. **MC noise:** small M for training, large M for evaluation; consider common random numbers.
4. **Sampling:** combine broad coverage with on-policy states.
5. **Residual choice:** Bellman residual is robust but may admit spurious solutions; Euler/KKT residuals sharpen optimality but require corner handling.
6. **Generalization:** test on-policy and stress-test outside training support.

3 Extension: Risky debt (Strebulaev and Whited (2012) Sec. 3.6-style)

This extension becomes a *joint fixed point*: equity value/policies interact with an endogenous pricing rule for risky debt.

3.1 Deep residual system (overview)

Networks. Use three MLPs:

$$V_{\theta_V}(k, b, z), \quad (k'_{\theta_P}(k, b, z), b'_{\theta_P}(k, b, z)), \quad \tilde{r}_{\theta_R}(z, k', b').$$

Bellman residual with limited liability. Let $\text{cont} = \text{flow} + \beta \mathbb{E}[V(k', b', z') \mid z]$ and impose limited liability:

$$V^{\text{target}} = \max(\text{cont}, 0).$$

Residual:

$$\mathcal{R}_{\text{bell}} = V_{\theta_V} - V^{\text{target}}.$$

Pricing residual (lender break-even). With soft default p_{def} and recovery proxy rec :

$$\mathcal{R}_{\text{price}}(k, b, z) = b'(1 + r) - \left(p_{\text{def}} \cdot \text{rec} + (1 - p_{\text{def}}) \cdot b'(1 + \tilde{r}) \right),$$

where $\tilde{r}$ is constrained with softplus so $1 + \tilde{r} > 0$.

Training loss.

$$\mathcal{L} = \mathbb{E}[\mathcal{R}_{\text{bell}}^2] + w_{\text{price}} \mathbb{E}[\mathcal{R}_{\text{price}}^2].$$

3.2 Numerical settings (risky debt experiment)

Please refer to Figure 2 and Table 3.2.

Item	Value / setting
Risk-free rate	$r = 0.02$ ($\beta \approx 0.980392$)
Technology / depreciation	$\theta = 0.7, \delta = 0.15$
Adjustment cost	$\phi_I = 0.01$
Taxes / default costs	$\tau = 0.2, \alpha = 0.2$
Shock parameters	$\rho = 0.7, \sigma_\varepsilon = 0.15$
State sampling	(k, z) log-uniform; $b \sim U[-20, 50]$
Networks	2-layer MLPs (256, 256) with tanh
Training	Adam, lr 2×10^{-4} , batch 4096, steps 10,000
MC draws	train 16; evaluation 512
Loss weight	$w_{\text{price}} = 0.2$

Table 3: Risky-debt DRM settings.

3.3 Interpreting typical training patterns

Common observations:

- Bellman residual falls faster than pricing residual (pricing is kinked and default-driven).
- Persistent $\mathbb{E}[p_{\text{def}}] \approx 0.38\text{--}0.40$ can be structural or due to state sampling mass in distress regions.
- Rising mean $\tilde{r}$ is consistent with high implied default risk and limited recovery.

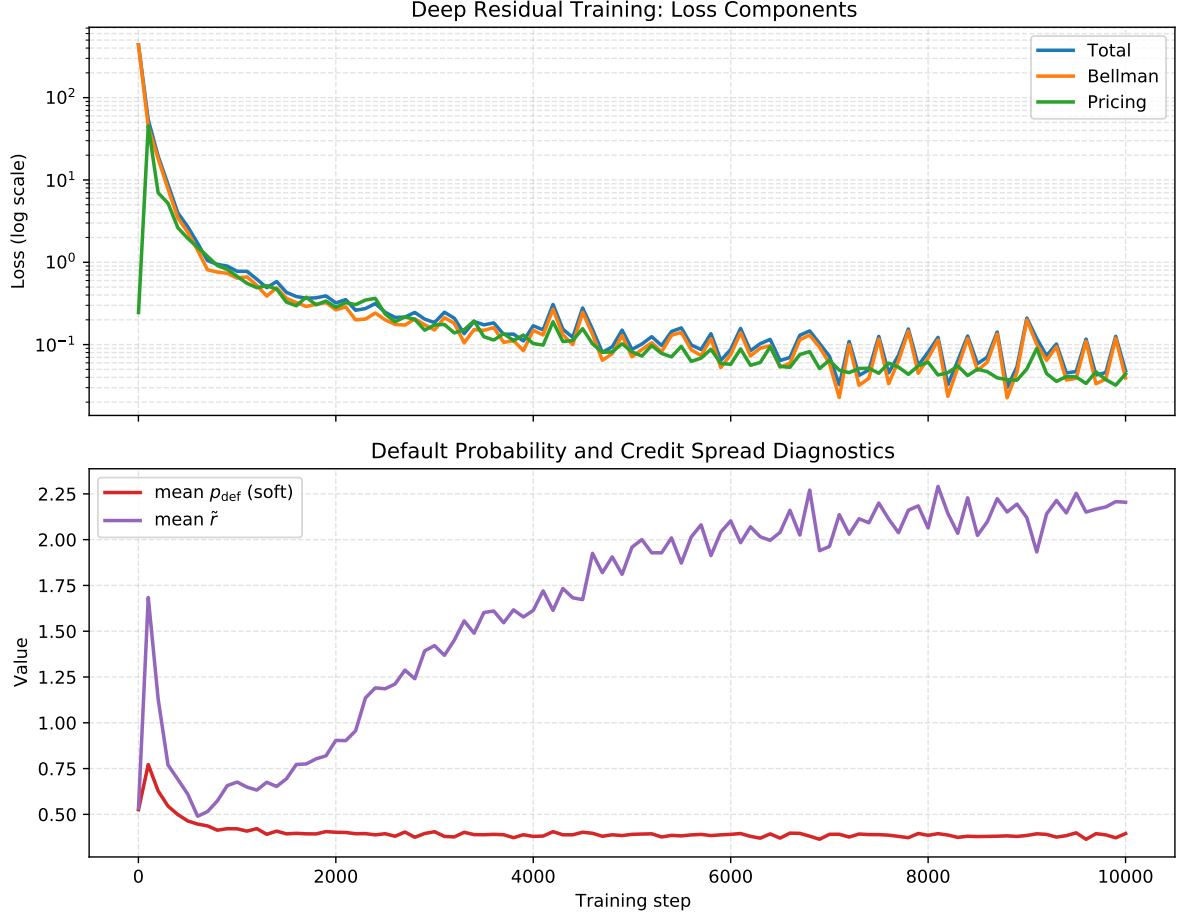


Figure 2: Training diagnostics parsed from console output. Top: total, Bellman, and pricing losses (log scale). Bottom: mean soft default probability and mean endogenous spread $\tilde{r}$.

4 Part 2. Estimation and application (GMM vs. SMM)

4.1 Estimators (compact definitions)

GMM. Let $g_t(\theta)$ be K moment conditions and $\hat{m}(\theta) = \frac{1}{T} \sum_{t=1}^T g_t(\theta)$. Then

$$\hat{\theta}_{\text{GMM}} \in \arg \min_{\theta \in \Theta} \hat{m}(\theta)^\top W \hat{m}(\theta).$$

SMM. Match empirical moments s_{data} with simulated $s_{\text{sim}}(\theta)$:

$$\hat{\theta}_{\text{SMM}} \in \arg \min_{\theta \in \Theta} (s_{\text{sim}}(\theta) - s_{\text{data}})^\top W (s_{\text{sim}}(\theta) - s_{\text{data}}).$$

4.2 Targets and results

Targets (example):

$$m_{\text{data}} = \begin{bmatrix} \text{default rate} \\ \text{mean spread} \\ \text{mean leverage} \end{bmatrix} = \begin{bmatrix} 0.4592 \\ 0.89526486 \\ 0.54685575 \end{bmatrix}, \quad g(\theta) = m_{\text{data}} - m_{\text{model}}(\theta).$$

Method	$J(\hat{\theta})$	Runtime (s)	g_1 (default)	g_2 (spread)	g_3 (lev.)	$\ g\ _2$
GMM	397.5	4.08	-0.1460	-0.0040	-0.0117	0.1465
SMM	139.9	21.2	+0.0760	-0.00145	-0.00137	0.0760

Table 4: GMM vs. SMM: reported objective, runtime, and moment errors.

Interpretation. SMM fits moments better in this run (smaller $\|g\|_2$) but is slower, consistent with simulation noise and higher inner-loop cost. Default-rate fit is the dominant source of mismatch.

5 Part 3. Bonus: CEO–Board incentives with effort, manipulation, and risk

5.1 Model block

Agents and controls. Board chooses contract parameters $c = (s_0, s_1, s_2)$. CEO chooses $a = (e, m, s)$: effort $e \geq 0$, manipulation $m \in \mathbb{R}$, risk $s \in [0, s_{\max}]$. Firm chooses k' (and b' under risky debt).

Reporting and risk. One implementable specification:

$$\pi^{\text{true}}(k, z, e) = zk^\theta + \chi_e e, \quad (11)$$

$$y = \log(\pi^{\text{true}}(k, z, e) + \varepsilon_0) + m + \eta, \quad \eta \sim \mathcal{N}(0, \sigma_y^2), \quad (12)$$

$$\log z' = \rho \log z + (\sigma_0 + s)\varepsilon', \quad \varepsilon' \sim \mathcal{N}(0, 1). \quad (13)$$

Costs and penalties.

$$C(e, m, s) = \frac{1}{2}\kappa_e e^2 + \frac{1}{2}\kappa_m m^2 + \frac{1}{2}\kappa_s s^2 + \text{Pen}(m), \quad (14)$$

$$p_{\text{det}}(m) = \frac{1}{1 + \exp(-\gamma m)}, \quad \text{Pen}(m) = \lambda p_{\text{det}}(m). \quad (15)$$

Compensation.

$$w(y; c) = s_0 + s_1 y + s_2 \max(y - \bar{y}, 0).$$

5.2 (a) Problems illustrated

1. Multi-dimensional moral hazard (effort vs. manipulation vs. risk).
2. Risk shifting under debt/limited liability and convex pay.
3. Misreporting affects financing terms if lenders condition on y .
4. Second-best contracting: board trades off incentives vs. distortions.

5.3 (b) How to modify [Strebulaev and Whited \(2012\) Sec. 3.1 / 3.6](#)

- Add CEO action block (e, m, s) and measurement equation for reported y .
- Add contract mapping $y \mapsto w(y)$.
- In risky debt: allow debt pricing/spreads to depend on y and default risk affected by s .

5.4 (c) Solving/estimating with CEO–Board

Approximate with DRM:

- $V_\theta(x)$, firm policies $k'(x)$ (and $b'(x)$), CEO best response $(e, m, s)(x, c)$, pricing $\tilde{r}(\cdot)$.
- Train on (i) Bellman residual, (ii) pricing residual, (iii) CEO-optimality residuals (FOC/KKT-style), plus feasibility penalties.
- Estimate via SMM/indirect inference matching both *real outcomes* (investment, leverage, default, spreads) and *reporting/contract* moments (pay sensitivities, convexity, misreporting proxies).

6 Training summary

Please refer to Figure 3.

6.1 Logged checkpoints (training-set diagnostics)

Bellman loss declines rapidly early in training (e.g. 3.651×10^2 at step 1 to 5.664×10^{-2} by step 6000). Pricing loss falls more slowly and remains material over checkpoints (e.g. 1.892 at step 500 to 6.471×10^{-2} at step 6000). CEO-FOC residual declines from 3.220×10^{-1} at step 1 to 5.265×10^{-2} at step 6000. One reading is that the model learns policies that stabilize default risk but the pricing block (or priced risk characteristics captured by $\tilde{r}$) evolves in a way that increases spreads, consistent with the pricing residual remaining non-negligible and/or the pricing network reweighting states during training.

6.2 Economic interpretation from checkpoint means

Mean manipulation rises over training ($m \approx -0.093$ to 1.222), mean effort falls ($e \approx 0.838$ to 0.112), and mean added risk declines modestly ($\sigma_{\text{add}} \approx 0.350$ to 0.210). Mean default probability declines toward ≈ 0.39 –0.41, while mean spread $\tilde{r}$ rises late in training (to ≈ 1.706 at step 6000).

6.3 Contract path

Contract parameters become more high-powered and convex:

$$(s_0, s_1, s_2) : (0.100, 0.200, 0.100) \rightarrow (0.162, 0.290, 0.185) \text{ by step 6000.}$$

In this run, the increase in incentive strength coincides with higher manipulation and lower effort, suggesting that (given the current penalty/detection technology and reporting specification) the contract loads more strongly on the reporting margin than on the real-effort margin.

6.4 Final evaluation metrics (out-of-sample)

- Bellman RMSE: 2.6670×10^{-1} ,
- Pricing RMSE: 2.3667×10^{-1} ,
- CEO FOC RMSE: 2.2808×10^{-1} ,
- mean default probability: 0.390,
- mean detection probability: $p_{\text{det}} = 0.851$,
- mean pay: 1.207,
- mean reported signal: $\bar{y}_{\text{rep}} = 2.170$.

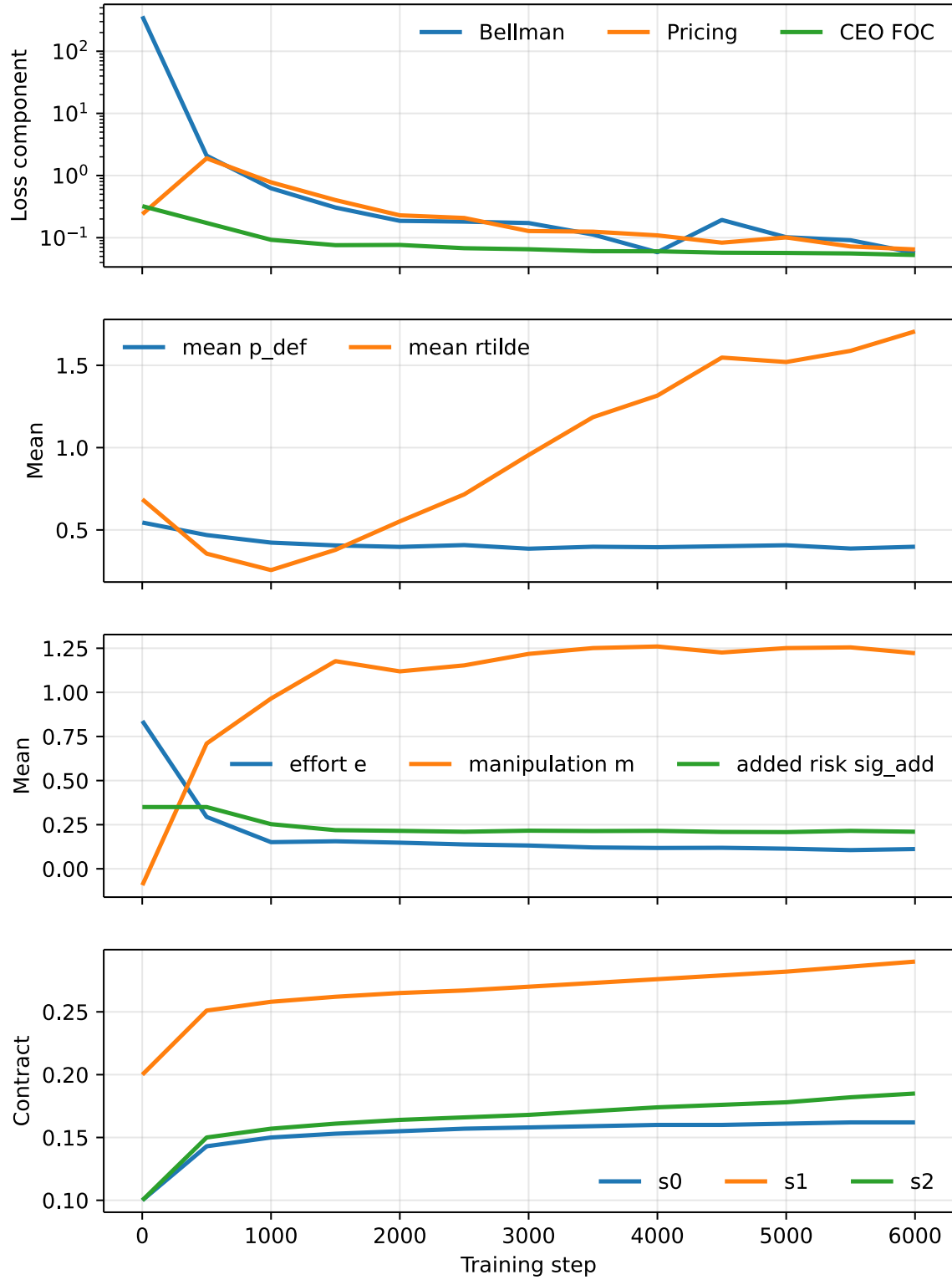


Figure 3: Training diagnostics (losses, credit outcomes, CEO actions, contract path)

Checkpoint vs. evaluation. Checkpoint metrics summarize the training distribution at sparse steps; evaluation RMSEs use a separate draw (and often higher MC), so gaps are informative about generalization and relative difficulty (typically pricing and CEO-optimality blocks).

7 Applications and limitations (brief)

Applications. Pricing corporate credit and decomposing spreads; leverage dynamics; stress testing; investment distortions; policy counterfactuals (tax/default cost changes).

Limitations / next steps. Richer debt maturity and rollover risk; more realistic default/recovery; aggregate risk and SDF-based pricing; improved measurement mapping and firm heterogeneity.

References

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